

화학수학

열 번째 수업

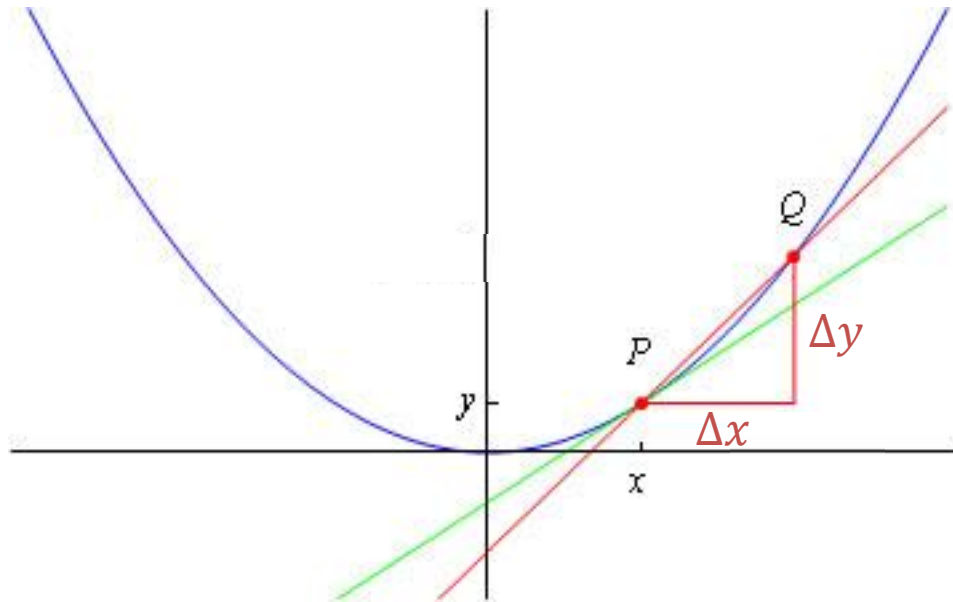
시험 안내 및 복습

중간고사

- 시간: 10월 21일(목) 오후 1시 반~2시 반
- 장소: 화학관 315호($\neq$ 교수계획표의 강의실)
- 계산기 불필요, 신분증 지참
- 10분 전까지 입실 완료
- 시험 범위: 다변수 미분 및 적분, 급수, 함수급수 및 적분변환
- 수업 시간에 다루지 않은 내용은 범위에 포함되지 않음

Differentiation

“The slope of a function”

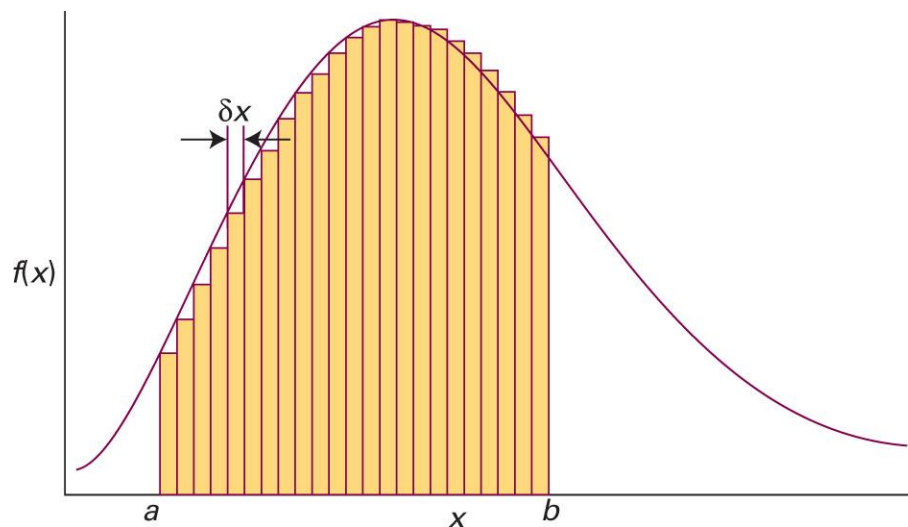


$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$

(Image: http://wwwf.imperial.ac.uk/metric/metric_public/differentiation/index.html)

Integration

“The areas under curves”



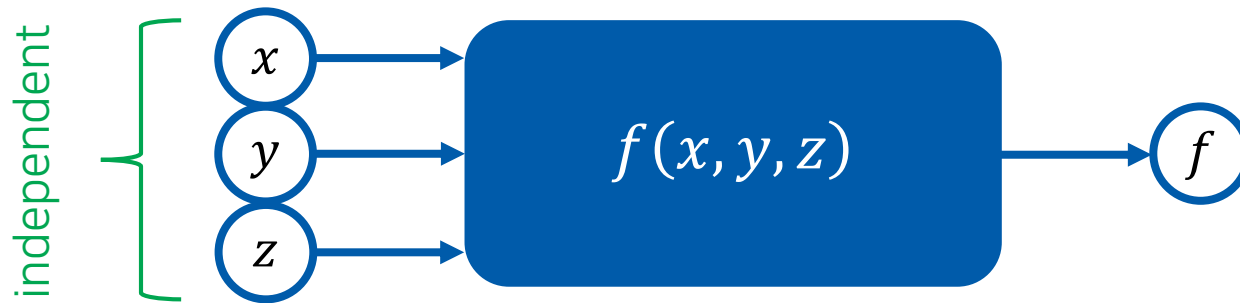
$$\lim_{\delta x \rightarrow 0} \sum_i f(x_i) \delta x = \int_a^b f(x) dx$$

8. Differential Calculus with Several Independent Variables

(8장 다변수 미분)

Function of Several Independent Variables

(다변수함수)



Example: $f(x, y, z) = x^2 + xy + 3z$

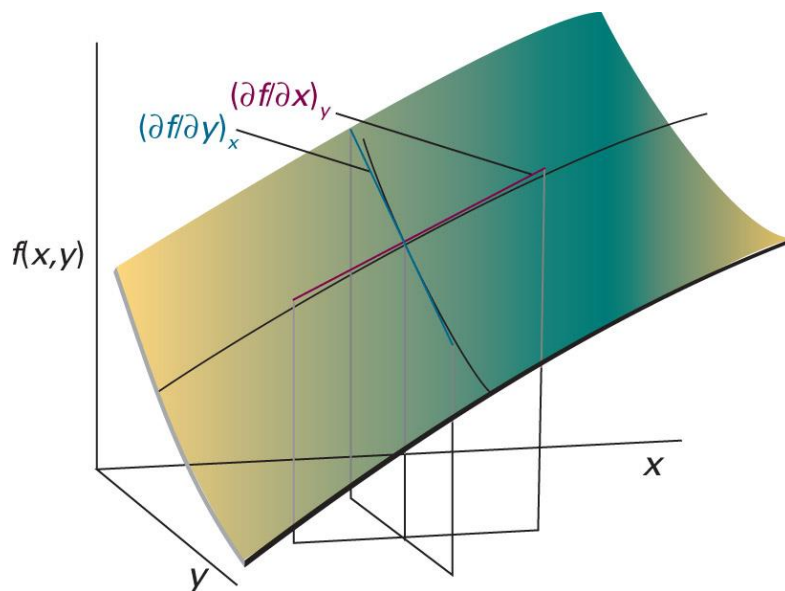
$$f(1, 1, 1) = 1 + 1 + 3 = 5$$

$$f(-1, 1, 0) = 1 - 1 + 0 = 0$$

⋮

Partial Derivatives (편미분)

“the slope of the function with respect to one of the variables,
all the other variables being held constant”



$$\left(\frac{\partial f}{\partial x}\right)_y = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\left(\frac{\partial f}{\partial y}\right)_x = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Fundamental Equation of Differential Calculus

(미분학의 기본 방정식)

- n independent variables: $x_1, x_2, \dots, x_n$
- dependent variable: y

The differential of y is

$$dy = \sum_{i=1}^n \left(\frac{\partial y}{\partial x_i} \right)_{x'} dx_i$$

where x' means keeping all of the variables except for x_i fixed

Change of Variables

We can use different sets of independent variables:

- $U = U(T, V, n)$

$$dU = \left(\frac{\partial U}{\partial T} \right)_{V,n} dT + \left(\frac{\partial U}{\partial V} \right)_{T,n} dV + \left(\frac{\partial U}{\partial n} \right)_{T,V} dn$$

- $U = U(T, P, n)$

$$dU = \left(\frac{\partial U}{\partial T} \right)_{P,n} dT + \left(\frac{\partial U}{\partial P} \right)_{T,n} dP + \left(\frac{\partial U}{\partial n} \right)_{T,P} dn$$

Be Careful

For most systems,

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} \neq \left(\frac{\partial U}{\partial T}\right)_{P,n}$$

Fixed variables (subscripts) are important!

Variable-Change Identity

(변수-변화 항등식)

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + \left(\frac{\partial U}{\partial P}\right)_{T,n} \left(\frac{\partial P}{\partial T}\right)_{V,n}$$

Reciprocal Identity

(역 항등식)

$$\left(\frac{\partial y}{\partial x}\right)_{z,u} = \frac{1}{(\partial x / \partial y)_{z,u}}$$

Euler Reciprocity Relation

(오일러 역관계식)

For $z = z(x, y)$,

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$$

Cycle Rule

(순환 법칙)

$$\left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = -1$$

Chain Rule

(연쇄 법칙)

For $z = z(u, x, y)$ and $x = x(u, v, y)$,

$$\left(\frac{\partial z}{\partial y}\right)_{u,v} = \left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v}$$

Note:

$$\left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v} \left(\frac{\partial y}{\partial z}\right)_{u,v} = 1$$

Check the subscripts!

Pfaffian Form: General Differential Form

(파피안형)

$$du = M(x, y)dx + N(x, y)dy$$

- Exact differential (완전 미분):

$$M(x, y) = \left(\frac{\partial u}{\partial x} \right)_y, \quad N(x, y) = \left(\frac{\partial u}{\partial y} \right)_x$$

- Inexact differential (불완전 미분): otherwise

Test of Exactness

$$du = M(x, y)dx + N(x, y)dy$$

Using the Euler reciprocity relation,

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

if the differential is exact,

$$\left(\frac{\partial N}{\partial x} \right)_y = \left(\frac{\partial M}{\partial y} \right)_x$$

Integrating Factor

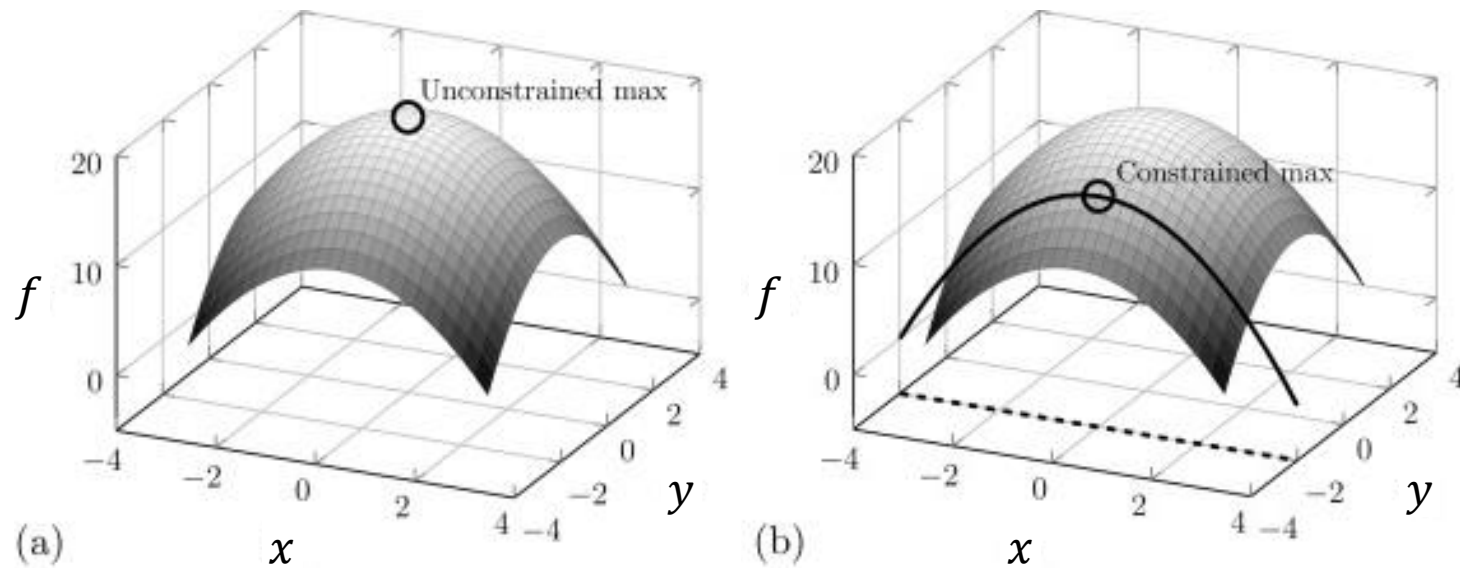
(적분인자)

inexact differential $\rightarrow$ exact differential
by multiplying a function (integrating factor)

No general method to find an integrating factor

Constrained Optimization

(제약 최적화)



with a constraint (제약 조건)

Lagrange's Method

Find the constrained extremum in $f(x, y)$,
when the constraint is written in the form $g(x, y) = 0$

1. New function $u(x, y) = f(x, y) + \lambda g(x, y)$
2. Partial derivatives of $u = 0$
3. Solve the equations from point 2 and $g(x, y) = 0$
to obtain x , y , and $\lambda \rightarrow$ constrained extrema

Vector Derivatives

- Gradient (기울기) ∇f

scalar $\rightarrow$ vector

- Divergence (발산) $\nabla \cdot \vec{F}$

vector $\rightarrow$ scalar

- Curl (회전) $\nabla \times \vec{F}$

vector $\rightarrow$ vector

- Laplacian (라플라스) $\nabla^2 f$

scalar $\rightarrow$ scalar (2nd order)

Vector Derivatives

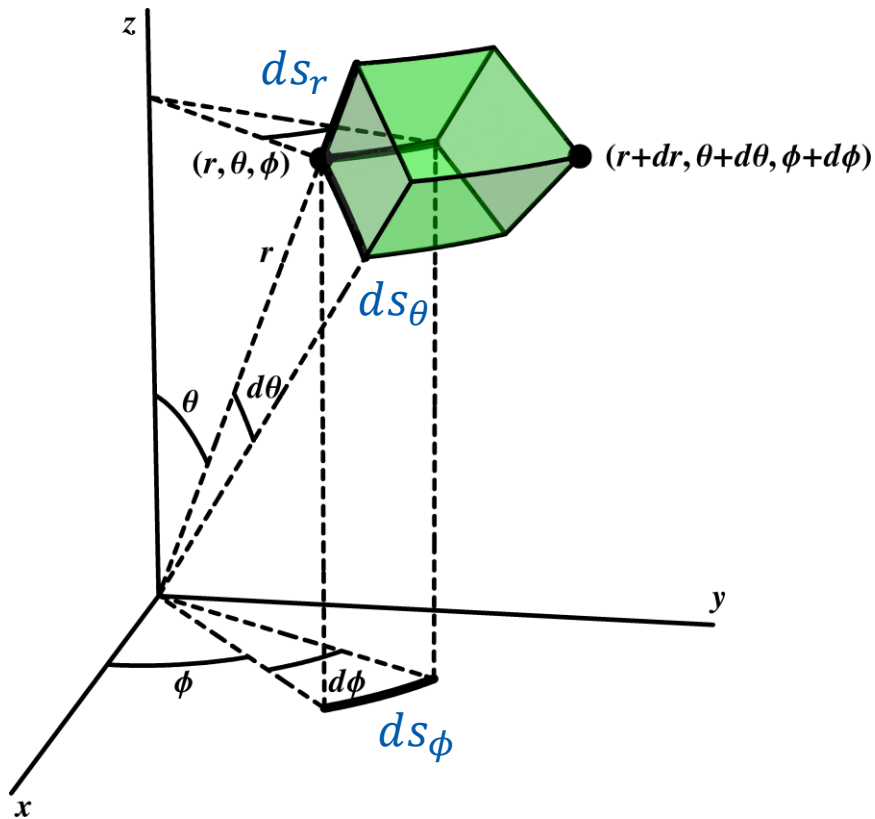
$$\nabla f = \hat{i} \left(\frac{\partial f}{\partial x} \right) + \hat{j} \left(\frac{\partial f}{\partial y} \right) + \hat{k} \left(\frac{\partial f}{\partial z} \right)$$

$$\nabla \cdot \vec{F} = \left(\frac{\partial F_x}{\partial x} \right) + \left(\frac{\partial F_y}{\partial y} \right) + \left(\frac{\partial F_z}{\partial z} \right)$$

$$\nabla \times \vec{F} = \hat{i} \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) + \hat{j} \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) + \hat{k} \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right)$$

$$\nabla^2 f = \left(\frac{\partial^2 f}{\partial x^2} \right) + \left(\frac{\partial^2 f}{\partial y^2} \right) + \left(\frac{\partial^2 f}{\partial z^2} \right)$$

Infinitesimal Displacements



$$ds_r = dr$$

$$ds_\theta = r d\theta$$

$$ds_\phi = r \sin \theta d\phi$$

Infinitesimal Displacements

In Cartesian coordinates,

$$d\vec{r} = \hat{i}dx + \hat{j}dy + \hat{k}dz$$

In spherical polar coordinates,

$$d\vec{r} = \hat{e}_r dr + \hat{e}_\theta r d\theta + \hat{e}_\phi r \sin \theta d\phi$$

In general,

$$d\vec{r} = \hat{e}_1 h_1 dq_1 + \hat{e}_2 h_2 dq_2 + \hat{e}_3 h_3 dq_3$$

Vector Derivatives in Other Coordinates

$$\nabla f = \hat{e}_1 \frac{1}{h_1} \frac{\partial f}{\partial q_1} + \hat{e}_2 \frac{1}{h_2} \frac{\partial f}{\partial q_2} + \hat{e}_3 \frac{1}{h_3} \frac{\partial f}{\partial q_3}$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial (F_1 h_2 h_3)}{\partial q_1} + \frac{\partial (F_2 h_1 h_3)}{\partial q_2} + \frac{\partial (F_3 h_1 h_2)}{\partial q_3} \right]$$

$$\begin{aligned} \nabla \times \vec{F} = & \hat{e}_1 \frac{1}{h_2 h_3} \left[\frac{\partial (h_3 F_3)}{\partial q_2} - \frac{\partial (h_2 F_2)}{\partial q_3} \right] + \hat{e}_2 \frac{1}{h_1 h_3} \left[\frac{\partial (h_1 F_1)}{\partial q_3} - \frac{\partial (h_3 F_3)}{\partial q_1} \right] \\ & + \hat{e}_3 \frac{1}{h_1 h_2} \left[\frac{\partial (h_2 F_2)}{\partial q_1} - \frac{\partial (h_1 F_1)}{\partial q_2} \right] \end{aligned}$$

Vector Derivatives in Other Coordinates

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_1 h_3}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

9. Integral Calculus with Several Independent Variables

(9장 다변수 적분)

Line Integral (선적분)

Consider a differential with two independent variables:

$$dF = M(x, y)dx + N(x, y)dy$$

For the curve C joining (x_0, y_0) and (x_1, y_1) ,
the **line integral** (path integral) is defined as

$$\int_C dF = \int_C [M(x, y)dx + N(x, y)dy]$$

Evaluation of Line Integral

$$\int_C dF = \int_C [M(x, y)dx + N(x, y)dy]$$

The curve can be expressed as $x = x(y)$ or $y = y(x)$, so

$$\int_C dF = \int_C [M\{x, y(x)\}dx + N\{x(y), y\}dy]$$

$$\int_C dF = \int_{x_0}^{x_1} M\{x, y(x)\} dx + \int_{y_0}^{y_1} N\{x(y), y\} dy$$

Line Integrals of Exact Differentials

If dF is an exact differential, the line integral $\int_C dF$

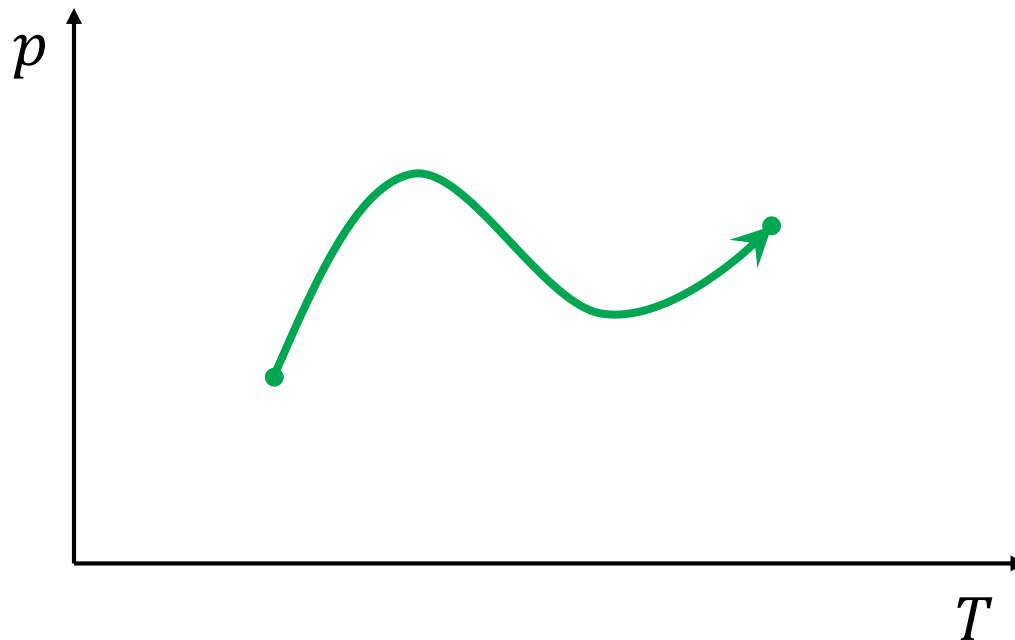
- depends **only on the initial and final points**.
- equals to **$F(\text{final point}) - F(\text{initial point})$** .

For an exact differential dF of the function $F(x, y)$,
with the initial point (x_0, y_0) and the final point (x_1, y_1) ,

$$\int_C dF = \int_C \left[\left(\frac{\partial F}{\partial x} \right)_y dx + \left(\frac{\partial F}{\partial y} \right)_x dy \right] = F(x_1, y_1) - F(x_0, y_0)$$

Line Integrals in Thermodynamics

- Equilibrium state: a point in a space
- Reversible process: a line integral



Double Integral (이중적분)

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} f(x, y) \, dy \, dx$$


- Carry out **the inner integration** first
 - Treat the other independent variable (x) as a constant
- Then carry out **the outer integration**

Triple Integral (삼중적분)

$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x, y, z) \, dz \, dy \, dx$$

- First, integrate z from c_1 to c_2 ($x, y = \text{constant}$)
- Then integrate y from b_1 to b_2 ($x = \text{constant}$)
- Lastly, integrate x from a_1 to a_2

Separation of Variables

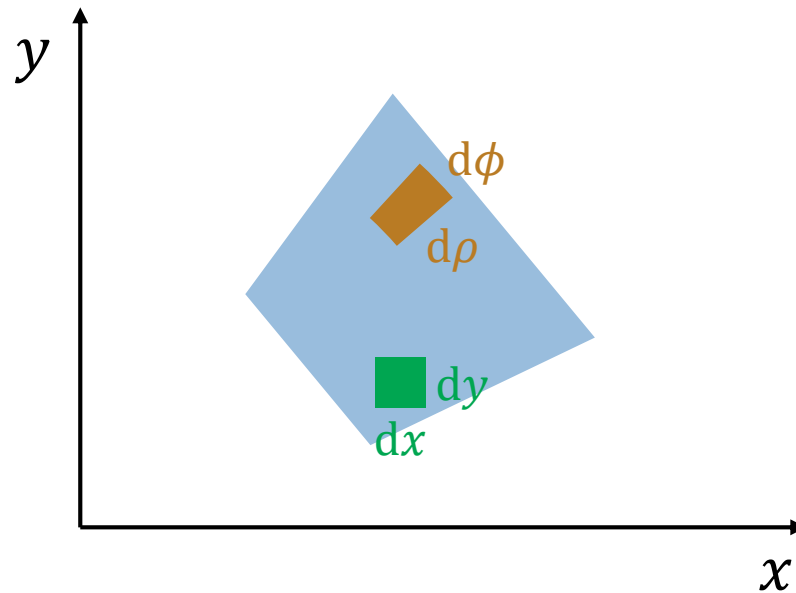
$$I = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x, y, z) \, dz \, dy \, dx$$

If $f(x, y, z) = f_x(x)f_y(y)f_z(z)$,

$$I = \left[\int_{a_1}^{a_2} f_x(x) \, dx \right] \times \left[\int_{b_1}^{b_2} f_y(y) \, dy \right] \times \left[\int_{c_1}^{c_2} f_z(z) \, dz \right]$$

In Different Coordinates

$$\iint f \, dA = \iint f(x, y) \, dx \, dy = \iint f(\rho, \phi) \rho \, d\rho \, d\phi$$



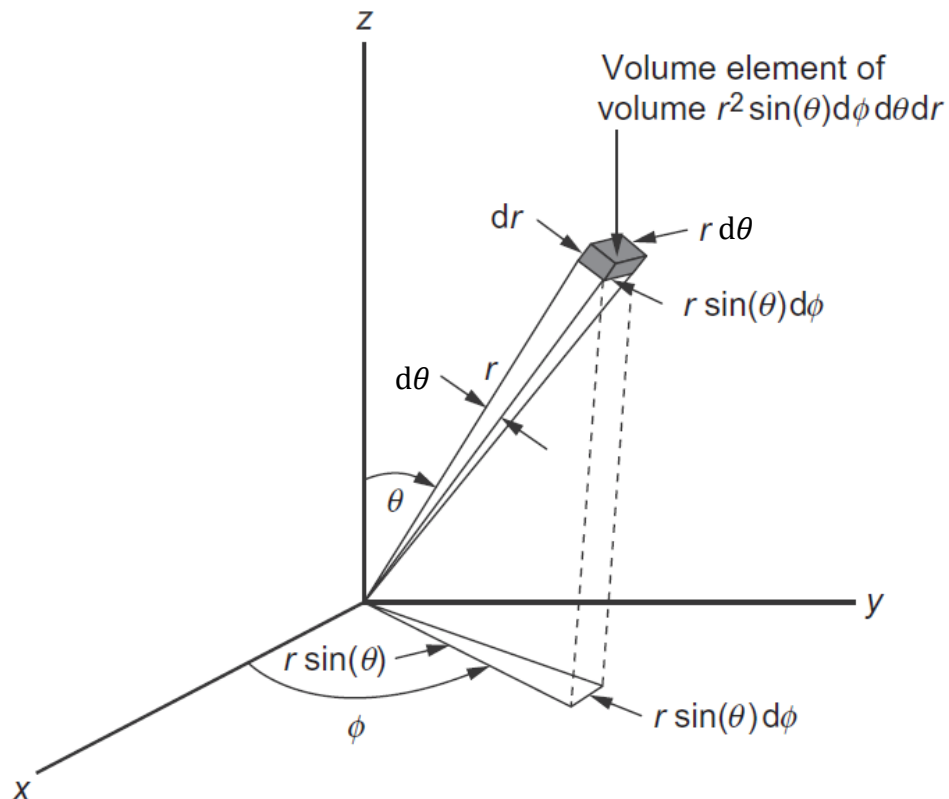
Jacobian

$$\int \int f(x, y) \, dx \, dy = \int \int f(\rho, \phi) \frac{\partial(x, y)}{\partial(\rho, \phi)} \, d\rho \, d\phi$$

where

$$\frac{\partial(x, y)}{\partial(\rho, \phi)} = \begin{vmatrix} \partial x / \partial \rho & \partial x / \partial \phi \\ \partial y / \partial \rho & \partial y / \partial \phi \end{vmatrix} = \left(\frac{\partial x}{\partial \rho} \right) \left(\frac{\partial y}{\partial \phi} \right) - \left(\frac{\partial x}{\partial \phi} \right) \left(\frac{\partial y}{\partial \rho} \right)$$

Triple Integral in Spherical Coordinates



Jacobian: $r^2 \sin \theta$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 9.5)

10. Mathematical Series

(10장 급수)

Sequence and Series

- **Sequence** (수열): $a_0 \rightarrow a_1 \rightarrow a_2 \rightarrow \cdots \rightarrow a_n \rightarrow \cdots$
 - Rule for generating one member from previous one
- **Series** (급수)
 - Finite series (유한급수): $s = a_0 + a_1 + \cdots + a_n$
 - Infinite series (무한급수): $s = a_0 + a_1 + \cdots + a_n + \cdots$

$$s = a_0 + a_1 + \cdots + a_n + \cdots = \sum_{k=0}^{\infty} a_k$$

Geometric Series (기하급수)

$$s = a + ar + ar^2 + ar^3 + \cdots + ar^n + \cdots = a \sum_{n=0}^{\infty} r^n$$

- Convergence condition: $|r| < 1$
- Value: since $s = a + r(a + ar + ar^2 + \cdots) = a + rs$,

$$s = \frac{a}{1 - r}$$

Functional Series (함수급수)

$$s(x) = a_0 g_0(x) + a_1 g_1(x) + a_2 g_2(x) + \cdots$$

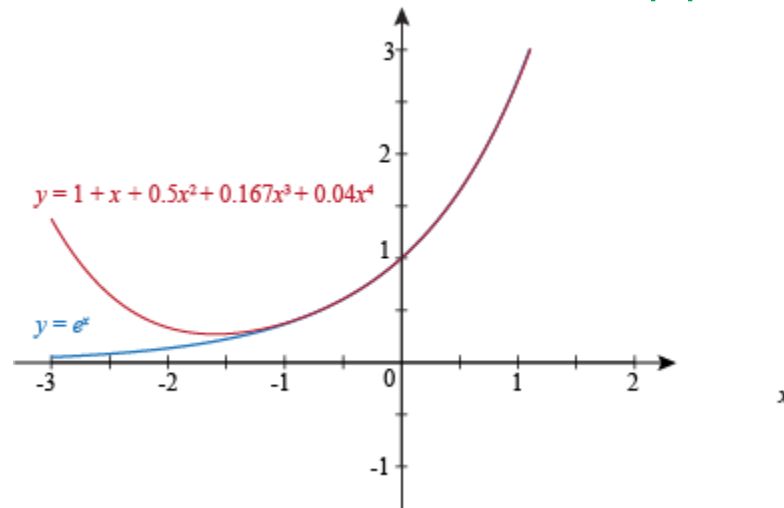
- Coefficients (계수): $a_0, a_1, a_2, \cdots$
 - Basis functions (바탕함수): $g_0(x), g_1(x), g_2(x), \cdots$
1. Power series (멱급수): powers as basis
 2. Fourier series (푸리에 급수): sines & cosines as basis

Approximation to a Function

- Functional series can be used to represent a function

$$s(x) = a_0g_0(x) + a_1g_1(x) + a_2g_2(x) + \cdots \rightarrow f(x)$$

- Its partial sum can be used as an approximation



(Image: <https://www.shmoop.com/series/taylor-maclaurin-series-examples.html>)

Maclaurin Series to Represent $f(x)$

$$f(x) = a_0 + a_1x + a_2x^2 + \dots$$

At $x = 0$, $f(0) = a_0$ and

$$\left(\frac{d^n f}{dx^n} \right)_{x=0} = n! a_n$$

Hence,

$$a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=0}$$

Taylor Series to Represent $f(x)$

$$f(x) = a_0 + a_1(x - h) + a_2(x - h)^2 + \dots$$

where $h \neq 0$.

$$a_0 = f(h), \quad a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=h}$$

- Often called **Taylor expansion about $x = h$**
- A function that is not differentiable at $x = 0$ can be represented by a Taylor series

Change of Variable

$$f(x) = a_0 + a_1(x - h) + a_2(x - h)^2 + \dots$$

can be written as

$$f(x + h) = a_0 + a_1x + a_2x^2 + \dots$$

$$a_0 = f(x = 0), \quad a_n = \frac{1}{n!} \left(\frac{d^n f}{dx^n} \right)_{x=0}$$

e.g. Taylor expansion of $\ln(x + 1)$:

$$\ln(x + 1) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \dots$$

Mathematical Operations on Series

Differentiation/integration of a series?

For a (uniformly convergent) functional series

$$f(x) = \sum_{n=0}^{\infty} a_n g_n(x),$$

$$\frac{df(x)}{dx} = \frac{d}{dx} \left[\sum_{n=0}^{\infty} a_n g_n(x) \right] = \sum_{n=0}^{\infty} a_n \frac{dg_n(x)}{dx}$$

$$\int_b^c f(x) dx = \int_b^c \left[\sum_{n=0}^{\infty} a_n g_n(x) \right] dx = \sum_{n=0}^{\infty} a_n \int_b^c g_n(x) dx$$

11. Functional Series and Integral Transforms

(11장 함수급수와 적분변환)

Fourier Series

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=0}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

- **Periodicity:** $f(x) = f(x + 2L) = f(x + 4L) = \dots$
- **Completeness:** the series can exactly represent any function

Orthogonality

The basis functions are orthogonal to each other:

$$\int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx = \begin{cases} 2L, & \text{if } m = n = 0 \\ L\delta_{mn}, & \text{otherwise} \end{cases}$$

$$\int_{-L}^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = \begin{cases} 0, & \text{if } m = n = 0 \\ L\delta_{mn}, & \text{otherwise} \end{cases}$$

$$\int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = 0$$

Fourier Series Coefficients

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) \, dx$$

$$a_{n \neq 0} = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) \, dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) \, dx$$

Fourier Series Coefficients

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

	Odd $f(x)$	Even $f(x)$
$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$	0	
$a_{n \neq 0} = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$	0	
$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$		0

sine series cosine series

Complex Exponential Basis Functions

$$a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) = \frac{a_n - ib_n}{2} e^{in\pi x/L} + \frac{a_n + ib_n}{2} e^{-in\pi x/L}$$

The following two formulas are equivalent:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}$$

Orthogonality and Coefficients

- Orthogonality

$$\int_{-L}^L (e^{im\pi x/L})^* e^{in\pi x/L} dx = 2L\delta_{mn}$$

- Coefficients

$$c_m = \frac{1}{2L} \int_{-L}^L f(x) e^{-im\pi x/L} dx$$

General Functional Series

- Complete set of orthogonal basis functions:

$$\psi_1, \psi_2, \dots, \psi_n, \dots$$

- Expansion of an arbitrary function f :

$$f = \sum_n c_n \psi_n$$

- Hilbert space: $\psi_1, \psi_2, \dots, \psi_n, \dots$ span the Hilbert space

Scalar Products in the Hilbert Space

- The scalar product of $f(x)$ and $g(x)$ is

$$\int f^*(x)g(x) dx$$

- The scalar product of $f(x, y, z)$ and $g(x, y, z)$ is

$$\iiint f^*(x, y, z)g(x, y, z) dx dy dz$$

Orthogonality and Normalization

- The basis functions are **orthogonal**:

$$\int \psi_n^* \psi_m \, dx = c \delta_{nm}$$

- They can be **normalized**:

$$\int \psi_n^* \psi_m \, dx = \delta_{nm}$$

- We assume the normalized basis functions

Finding the Coefficients

$$f(x) = \sum_n c_n \psi_n$$

Multiplying both sides by ψ_m^* and integrating,

$$\int \psi_m^* f dx = \sum_n c_n \int \psi_m^* \psi_n dx$$

$$\int \psi_m^* f dx = \sum_n c_n \delta_{mn} = c_m$$

Integral Transforms



Fourier transform

Laplace transform

Mellin transform

Hankel transform

⋮

Transform and Inverse Transform

- Fourier transform

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx$$

- Inverse Fourier transform

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(k) e^{ikx} dk$$

Fourier Cosine and Sine Transforms

- Fourier cosine transform: for an **even** function $f(x)$,

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \cos(kx) dx$$

- Fourier sine transform: for an **odd** function $f(x)$,

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \sin(kx) dx$$

One-Sided Transforms

- One-sided Fourier transform ($x > 0$)

$$F(k) = \left(\frac{2}{\pi}\right)^{1/2} \int_0^{\infty} f(x) e^{-ikx} dx$$

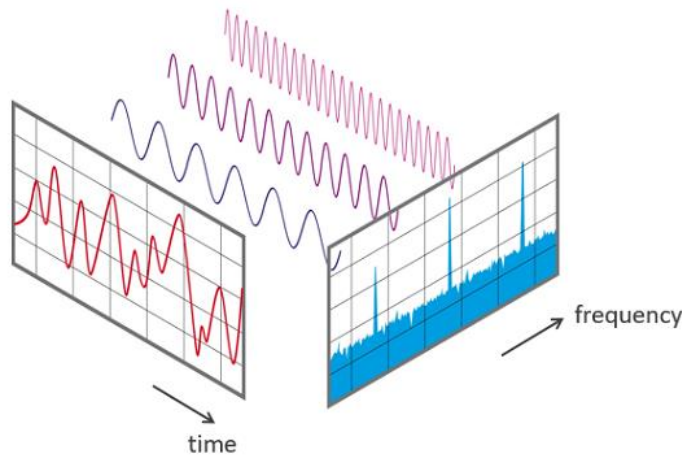
- One-sided inverse Fourier transform ($k > 0$)

$$f(x) = \left(\frac{2}{\pi}\right)^{1/2} \int_0^{\infty} F(k) e^{ikx} dk$$

Application: Time $\leftrightarrow$ Frequency Transform

$$F(\omega) = \left(\frac{2}{\pi}\right)^{1/2} \int_0^{\infty} f(t) e^{-i\omega t} dt$$

$$f(t) = \left(\frac{2}{\pi}\right)^{1/2} \int_0^{\infty} F(\omega) e^{i\omega t} d\omega$$



(Image: <https://www.nti-audio.com/en/support/know-how/fast-fourier-transform-fft>)

Some Common Fourier Transforms

$f(x)$	$F(k)$	$f(x)$	$F(k)$
1	$\sqrt{2\pi}\delta(k)$	$e^{-a x }$	$\left(\frac{2a^2}{\pi}\right)^{1/2} \frac{1}{k^2 + a^2}$
e^{iax}	$\sqrt{2\pi}\delta(k - a)$	$xe^{-a x }$	$-\left(\frac{2a^2}{\pi}\right)^{1/2} \frac{2ik}{(k^2 + a^2)^2}$
e^{-iax}	$\sqrt{2\pi}\delta(k + a)$	$e^{-a^2x^2}$	$\frac{1}{\sqrt{2}a} e^{-k^2/4a^2}$
$1/x$	$-i\left(\frac{\pi}{2}\right)^{1/2} \text{sgn}(k)$	$\frac{1}{x^2 + a^2}$	$\left(\frac{\pi}{2a^2}\right)^{1/2} e^{-a k }$

- Dirac delta function $\delta(x) = \begin{cases} \infty & \text{if } x = 0 \\ 0 & \text{if } x \neq 0 \end{cases}$
- Sign function $\text{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$

Laplace Transforms

- Laplace transform

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt = \mathcal{L}\{f(t)\}$$

- Inverse Laplace transform

$$f(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s)e^{st} ds = \mathcal{L}^{-1}\{F(s)\}$$

requires complex analysis;
We will not use this formula

Some Common Laplace Transforms

$F(s)$	$f(t)$
$1/s$	1
$1/s^2$	t
$n!/s^{n+1}$	t^n
$\frac{1}{s-a}$	$e^{at} (s > a)$
$\frac{s}{s^2 + a^2}$	$\cos (at)$
$\frac{a}{s^2 + a^2}$	$\sin (at)$
$\frac{s}{s^2 - a^2}$	$\cosh (at)$
$\frac{a}{s^2 - a^2}$	$\sinh (at)$

Useful Theorems

- Shifting theorem

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a)$$

- Derivative theorem

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = s\mathcal{L}\{f(t)\} - f(0)$$

- Integral theorem

$$\mathcal{L}\left\{\int_0^t f(u) du\right\} = \frac{1}{s}\mathcal{L}\{f(t)\}$$